EDGE AI WITH THE RASPBERRY PI

Course Roadmap: From Absolute Beginner to Edge AI Engineer

46 core days of about 2 hours each (roughly nine weeks at five days a week), on a Raspberry Pi 4 or Raspberry Pi 5, plus an optional Arduino UNO Q track.

The course in one picture

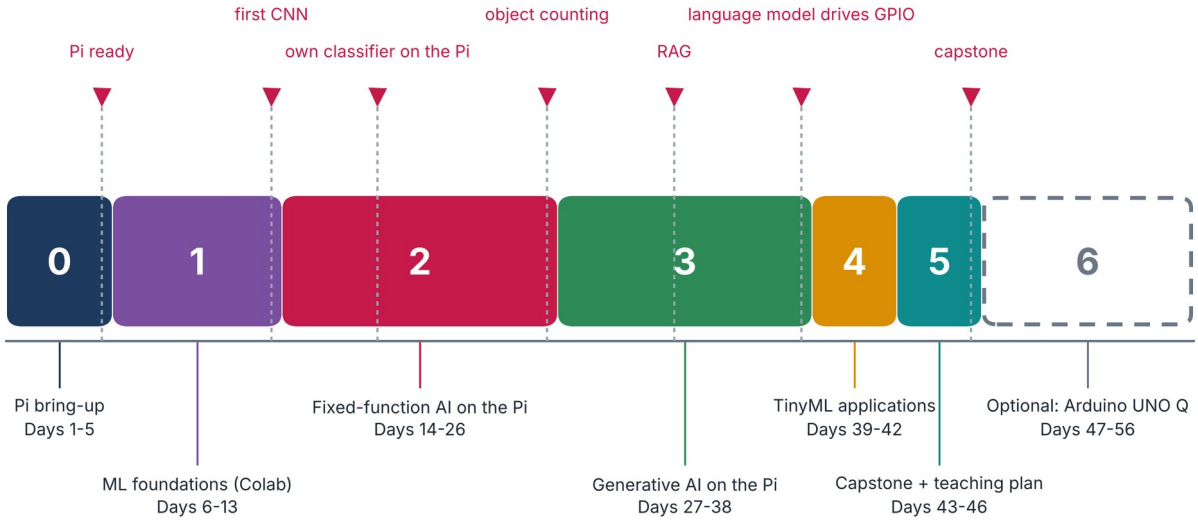


Figure 1. Six phases, 46 core days, and an optional extension. The red flags are milestones you will be able to demonstrate.

Phase	Days	What you learn	What you need
0 Pi bring-up	1-5	Flash, boot, remote access, Linux basics, camera, Python and Jupyter on the Pi	Raspberry Pi 4 or 5, microSD, power supply, laptop
1 ML foundations (Colab)	6-13	Neurons, deep neural networks, CNNs, datasets, metrics, overfitting, Edge Impulse first steps	Laptop and Google Colab; the Pi is not needed
2 Fixed-function AI on the Pi	14-26	Image classification, object detection (SSD, FOMO), YOLO, counting objects	Pi camera or USB webcam; Edge Impulse account
3 Generative AI on the Pi	27-38	RNN text, distillation, small language models, RAG, tools, agents, vision-language, audio, GPIO, IoT control	USB microphone and speaker; breadboard, LEDs, button, sensors
4 TinyML applications	39-42	Anomaly detection, keyword spotting, image applications on microcontrollers, responsible AI	Concepts only: no microcontroller board needed
5 Capstone + teaching plan	43-46	Your own project, the write-up, and the plan to teach it to students	Whatever the project needs
6 Optional: Arduino UNO Q	47-56	The same ideas on a second hardware platform, after the Pi and ML are mastered	Arduino UNO Q board (optional)

How the course works

- **One day is about two hours.** Each day has a full tutorial (complete in itself: no book or website is needed), a narrated animation where it helps, hands-on steps, a quiz and a set of notebook-style notes with space for your own comments.
- **Pi 4 and Pi 5.** Every day works on both boards. Where the boards differ (speed, memory, power, camera cable, model sizes) the lesson says so.

- **Learn it twice.** If you have two Pis, do each day once on one Pi (the learning pass), then repeat it on the other from your notes only (the consolidation pass). Give the Pis different host names, keep the same user name, and keep your code and notes in a shared online repository.
- **Phase 1 needs no Pi:** only Google Colab, so it can be done anywhere.
- **Lesson sizes** are estimates. A day that needs longer is simply continued the next day.

CAREFUL **Notes on hardware and software**

- **Pi 4 and Pi 5 speed and memory:** the vision phase runs well on both. In the language-model phase a Pi 5 (8 GB is best) runs larger models; a Pi 4 uses smaller models and longer waits. Each lesson recommends a model size per board.
- **Software versions:** a newer Raspberry Pi OS may ship a newer Python than some machine-learning packages support. Days 4 and 5 include the check and the remedy (virtual environments, or an older OS image).
- **Online services** (Colab, Edge Impulse, model downloads) need Internet access; if your network restricts them, use a phone hotspot on those days.
- **Arduino UNO Q** is an optional extension; the core course does not need it.

What to prepare, and when

By day	Item	Why
1	Raspberry Pi 4 or 5, microSD (32 GB or more), the matching USB-C power supply, card reader, Wi-Fi details	Everything in Phase 0
4	Pi camera module (check the camera cable: it differs between Pi 4 and Pi 5) or a USB webcam	Camera test and all vision projects
6	Google account	Colab, ML foundations
13	Edge Impulse account (check the current free tier)	Training in Phase 2
27	Internet on the Pi for model downloads; enough free space on the card	Ollama and models
35	USB microphone and speaker	Audio pipeline
36	Breadboard, jumper wires, LEDs with resistors, push button, DHT temperature sensor, BMP280 pressure sensor (confirm the exact parts in the lessons before buying)	GPIO and sensors
47+	Arduino UNO Q (optional)	Alternative hardware track

Day-by-day syllabus

Day	Topic	What you do	You will have
Phase 0 - Pi bring-up (days 1-5)			
1	Meet the Raspberry Pi; flash the microSD; first boot and SSH	Learn Edge AI and the Pi 4 / Pi 5; write the OS to a card; boot; log in from your laptop	A Pi you can log into remotely
2	Remote access in depth	Copy files both ways (scp, SFTP); open the desktop remotely; add a second Wi-Fi network; work with more than one Pi	Files and desktop reachable from your laptop
3	The Linux command line	Navigate, create, edit and delete files; permissions; processes; monitoring with htop	Confidence in the terminal
4	Camera, swap memory and updates	Connect a Pi camera or USB webcam; capture a photo; adjust swap; update the system	First photo captured on the Pi
5	Python environment and Jupyter on the Pi	Create a virtual environment; install packages; run Jupyter remotely	A notebook running on the Pi
Phase 1 - ML foundations (Colab) (days 6-13)			
6	Google Colab, Python refresher, what TinyML is	Run notebooks in the cloud; revise Python and NumPy	A working Colab setup
7	The machine-learning paradigm	Fit a first model from data and compare it with rule-based code	Your first trained model
8	Neural-network building blocks	Neurons, layers, activations, loss, gradient descent; build a small network	A small network trained from scratch
9	Regression with a deep neural network	Predict a number from data; plot predictions	A regression model
10	Classification with a deep neural network	Predict classes; read a confusion matrix	A classifier with metrics
11	Datasets, splits and metrics	Train / validation / test splits; accuracy, precision, recall, F1	Correct model evaluation
12	Convolutions; image classification with a CNN	Filters and pooling; build and train a CNN	A CNN trained in Colab
13	Edge Impulse first project; preventing overfitting	Train in the browser; dropout and data augmentation	A model ready to export
Phase 2 - Fixed-function AI on the Pi (days 14-26)			
14	TensorFlow Lite and the first inference on the Pi	Install TensorFlow Lite; run a pre-trained image classifier	An image classified on the Pi
15	Live camera classification; a model trained from scratch	Classify live camera frames; run a small CIFAR-10 model	Live classification
16	Custom classifier: goal and data collection	Define the problem; collect and organise your own images	Your own dataset
17	Training in Edge Impulse	Impulse design, pre-processing, training, accuracy versus speed	A trained, quantised model
18	Test, deploy and run live on the Pi	Test the model; deploy it; run live inference	Your own classifier on the Pi
19	Object detection fundamentals	Bounding boxes, labels, post-processing; a pre-trained detector	Detection on still images
20	Detection on a live stream	Run a detector on the camera stream	Live object detection
21	Custom detection: data and labelling	Collect and label images for detection	A labelled detection dataset
22	SSD MobileNet in Edge Impulse; deployment	Train, deploy and post-process an SSD model	A detector on the Pi
23	FOMO: lightweight detection	Train and deploy a FOMO model	FOMO running on the Pi
24	YOLO with Ultralytics on the Pi	Run YOLO; export to an efficient format	YOLO inference on the Pi
25	YOLO: custom training and deployment	Train YOLO on your data; deploy it	A custom YOLO model
26	Counting objects; Phase 2 mini-project	Build a counting pipeline; complete a small project	A counting system

Day	Topic	What you do	You will have
Phase 3 - Generative AI on the Pi (days 27-38)			
27	Text generation with an RNN	Tokens, embeddings, training a small text generator	A tiny text generator
28	Knowledge distillation	Teach a small student model from a larger teacher	A distilled model
29	Small language models: first model with Ollama	Install Ollama; choose a model that fits your board; chat	A language model answering on the Pi
30	Ollama with Python	Use the Python library; build a chat script	A Python chat program
31	Function (tool) calling	Let the model call Python functions	A tool-using model
32	Retrieval-augmented generation (RAG)	Build a vector store; retrieve context for answers	Question answering over your documents
33	Agents with small models	Design a tool set; build a simple agent loop	A working agent
34	Vision-language models	Run a vision-language model (Florence-2) for captioning and detection	Image captioning on the Pi
35	Audio and vision AI pipeline	Microphone to speech-to-text to language model to speech	Speech in, answer out
36	GPIO and physical computing: LEDs and buttons	Wire a circuit; control it from Python	An LED and button under code control
37	Sensors and notebooks	Read temperature, humidity and pressure sensors; log data	Sensor data logged
38	Small language models for IoT control	Let a language model switch real hardware	Language-model-driven hardware
Phase 4 - TinyML applications (days 39-42)			
39	Anomaly detection and K-means	Cluster data; detect anomalies	An anomaly detector (concept + notebook)
40	Keyword spotting (concepts)	Audio features and the keyword-spotting pipeline	The pipeline understood
41	Image applications on microcontrollers; data collection	Person detection concept; ways to collect data	A data-collection plan
42	Responsible AI; going further with Edge AI	Fairness, privacy, safety, lifecycle	A responsible-deployment checklist
Phase 5 - Capstone + teaching plan (days 43-46)			
43	Choose and plan the project	Define the goal, data, model and hardware	A one-page project plan
44	Build	Implement the system	A working prototype
45	Build, test and measure	Measure accuracy, speed and resource use honestly	Measured results
46	Present, write up, plan how to teach it	Report, demo, lesson plan	Report + teaching plan
Phase 6 - Optional: Arduino UNO Q (days 47-56)			
47	UNO Q: foundations and setup	Understand the board; set it up	Board working
50	Language models on the UNO Q; benchmarking; Python	Run and benchmark models; script them	Model running
53	Multimodal and agentic AI on the UNO Q	Combine vision, language and tools	An agent running
55	Physical AI: the two processors working together	Connect a language model to hardware through the bridge	A language model driving hardware

NOTE Checkpoints: what you can demonstrate

- **Day 5:** a Pi you can log into from your laptop, with Python and Jupyter ready.
- **Day 13:** a convolutional network you trained yourself. **Day 18:** your own image classifier running live on the Pi.
- **Day 26:** objects counted by a vision system. **Day 32:** a question-answering system over your own documents (RAG).
- **Day 38:** a small language model switching real hardware. **Day 46:** your capstone project and the plan to teach it.

Acknowledgments and references

This course is a compilation: it combines material from the sources below with new explanations, examples, diagrams, exercises and comments written for this course. The original authors are gratefully acknowledged.

[1] Rovai, M. (2025). *Edge AI Engineering: Hands-on with the Raspberry Pi*. Open textbook for the course IESTI05, Federal University of Itajubá (UNIFEI), Brazil. Version of 23 October 2025. Online edition: https://mjrovai.github.io/EdgeML_Made_Ease_ebook/

[2] Rovai, M. (2026). *IESTI05 - Edge AI: Edge Machine Learning Systems Engineering*. Course slides, notebooks and video lectures (second semester 2026), UNIFEI, Brazil. Course videos listed by the author: Class 1 (course introduction) https://youtu.be/QAb3MFHwe78 ; Class 2 (Introduction to Linux) https://youtu.be/QkIHILtl0h4 ; Class 3 (Raspberry Pi setup walkthrough) https://youtu.be/YtjsM36LzVE (lectures in Portuguese).

[3] Rovai, M. (2026). *IESTI01 - TinyML: Machine Learning for Embedding Devices*. Course materials (first semester 2026), UNIFEI, Brazil. https://github.com/Mjrovai/UNIFEI-IESTI01-TinyML

[4] Rovai, M. *TinyML4D*: lectures, books, papers and tutorials for embedded machine-learning education. https://github.com/Mjrovai/TinyML4D

[5] Raspberry Pi Imager: download page and software. https://www.raspberrypi.com/software/

[6] Raspberry Pi Ltd. *Raspberry Pi documentation* (hardware specifications of the Raspberry Pi 4 and Raspberry Pi 5, operating system, remote access). https://www.raspberrypi.com/documentation/